

Algorithm for file updates in Python - SDR

Project description

This project demonstrates how repetitive tasks can be automated through python to save time and effort when performing security operations. The program iterates through a list of ip addresses and ensures that the list contains no entries found in a second “remove” list.

Open the file that contains the allow list

```
# Build `with` statement to read in the initial contents of the file  
with open(import_file, "r") as file:
```

Open ‘import_file’ with read permissions and assign to ‘file’ variable

Read the file contents

```
# Use `.read()` to read the imported file and store it in a variable named `ip_addresses`  
ip_addresses = file.read()
```

Use the .read() method to store ‘import_file’ in the ‘ip_addresses’ variable

Convert the string into a list

```
# Use `.split()` to convert `ip_addresses` from a string to a list  
ip_addresses = ip_addresses.split()
```

Split the long string into a clean list of ip addresses

Iterate through the remove list

```
for element in ip_addresses:

    # Build conditional statement
    # If current element is in `remove_list`,

    if element in remove_list:
```

Iterate through the ip_addresses list, checking if each element is also in 'remove_list'

Remove IP addresses that are on the remove list

```
    # then current element should be removed from `ip_addresses`

    ip_addresses.remove(element)
```

If an entry is included in both lists, remove it from the primary 'ip_addresses' list

Update the file with the revised list of IP addresses

```
# Convert `ip_addresses` back to a string so that it can be written into the text file
ip_addresses = " ".join(ip_addresses)

# Build `with` statement to rewrite the original file

with open(import_file, "w") as file:

    # Rewrite the file, replacing its contents with `ip_addresses`

    file.write(ip_addresses)
```

Update the desired list of ip addresses using open() with write permissions

Summary

This simple algorithm for comparing the two lists can easily be formatted as a simple function and called whenever necessary for the security analyst. This is just one example of a security compliance task that can be automated using basic python knowledge.